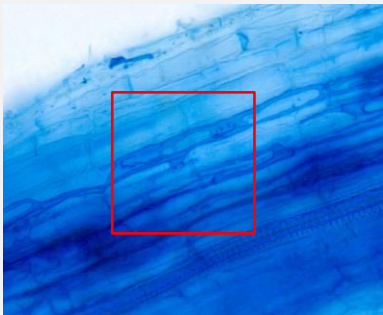
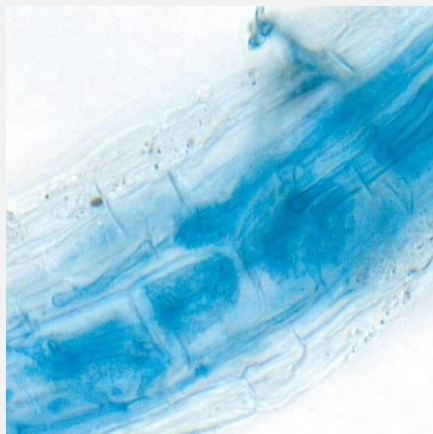
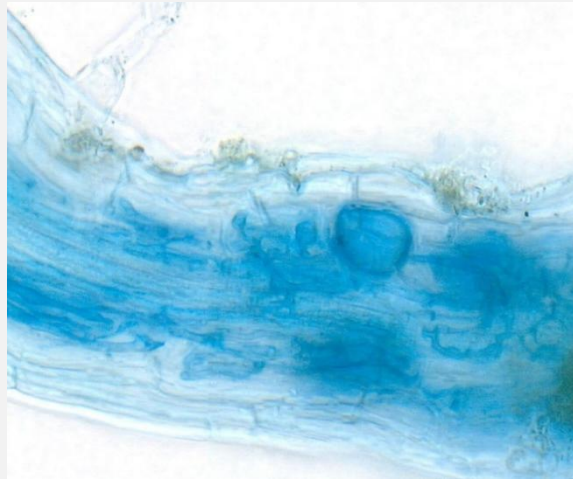
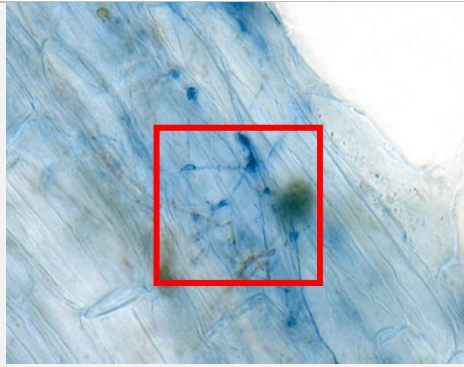


# MycorrhizaFinder AM Classification Guidelines

**General:** The manual labelling process should be guided by *clearly defined criteria* whilst being *intuitive*. Given a central image tile, experts should label as though assessing through a microscope given the context of the surrounding 8 tiles, while following the guidelines presented herein. This document aims to define *clear classification criteria* for all classes encountered, which can be used across numerous researchers to curate a new training dataset.

| Class                                            | Criteria                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|--------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| [AM+] Colonised root<br>(Arbuscular mycorrhizae) | <p><b>Must exhibit at least one of the following image features:</b></p> <ul style="list-style-type: none"><li>■ Intracellular Hyphae (without regular septa)</li><li>■ Vesicles if attached to intracellular AM hyphae</li><li>■ Arbuscules</li></ul> <p><b>Amount of colonisation required:</b><br/>No minimum requirement but hyphae must have both edges clearly visible within central tile</p> <p>Vesicles or arbuscules which cross the central tile must be identifiable within context of the 9 tiles visible.</p> <p><b>Exemplary images:</b></p>  |

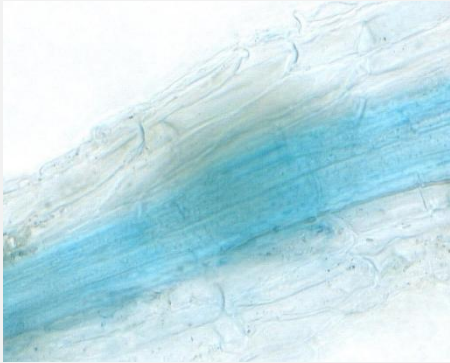
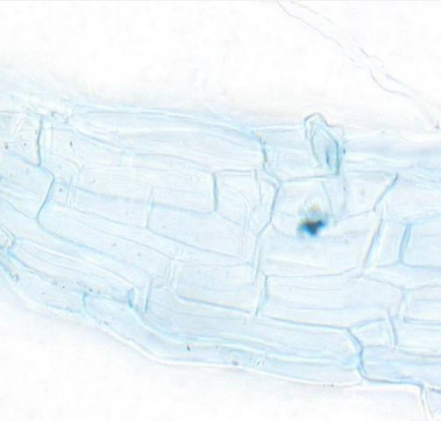


**Other aspects to be aware of:**

A clearly visible colonised root hair can be considered colonised root.

If regular septa are present, see DSE or Not Colonised

If DSE are present alongside AM hyphae, see Hybrid

|                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|--------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p>[ErM+] Colonised root<br/>(Ericoid mycorrhizae)</p> | <p><b>Must exhibit at least one of the following image features:</b></p> <ul style="list-style-type: none"> <li>▪ Intracellular coils (with regular septa)</li> </ul> <p><b>Amount of colonisation required:</b><br/>No minimum requirement but hyphae must have both edges clearly visible within central tile</p> <p><b>Exemplary images: See Ericoid Guideline Doc</b></p> <p><b>Other aspects to be aware of:</b><br/>DSE may also form coils in Ericoid roots, but is likely to stain a darker colour</p>               |
| <p>[N-] Non-Colonised root</p>                         | <p><b>Must exhibit following image features:</b></p> <ul style="list-style-type: none"> <li>■ Plant cell walls</li> <li>■ Lack of intracellular AM fungi</li> </ul> <p><b>Percentage of the tile occupied:</b><br/>10%</p> <p><b>Exemplary images:</b></p> <div data-bbox="635 1238 1086 1599">  </div> <div data-bbox="635 1608 1077 2027">  </div> |



**Other aspects to be aware of:**

A clearly visible root hair can count as root

May contain:

- Sclerotia, Bacteria, Unidentified septate fungi, etc.

[D] DSE

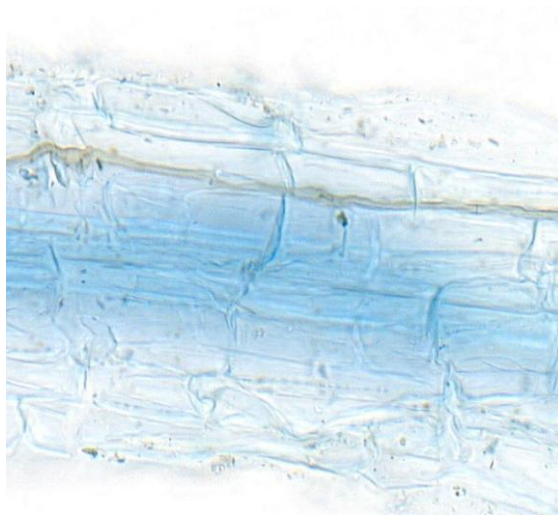
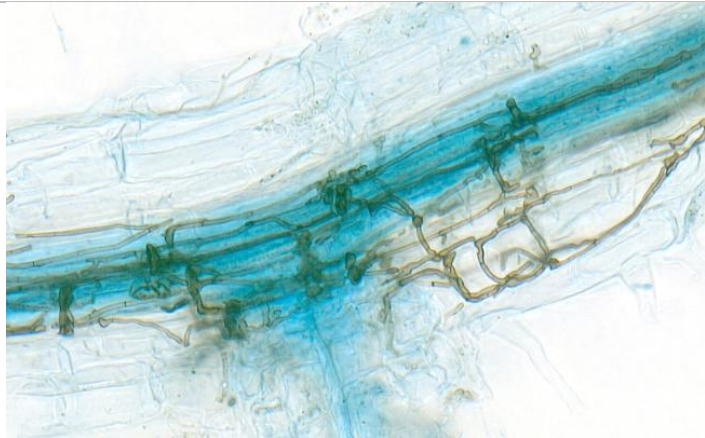
**Must exhibit one of the following image features:**

- Melanised Hyphae with regular septa
- DSE microsclerotia

**Percentage of the tile occupied:**

See rules used for AM fungi in AM+

**Exemplary images:**



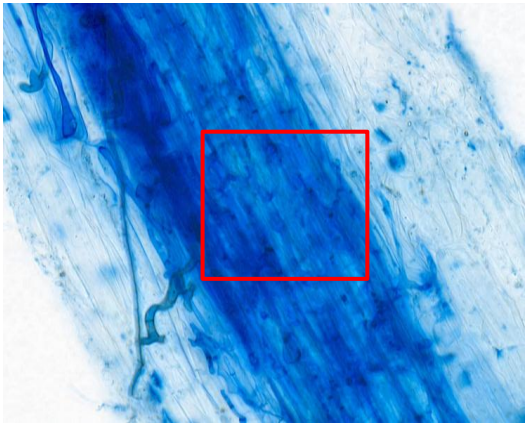
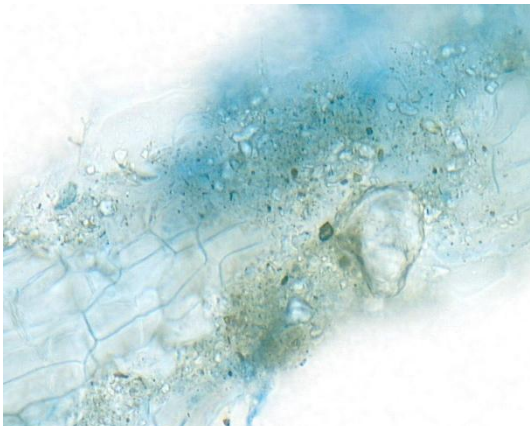
**Other aspects to be aware of:**

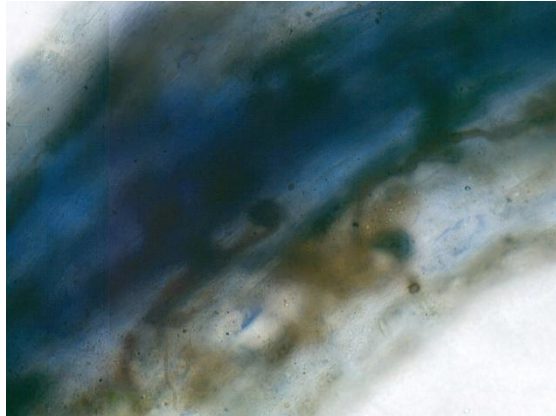
If DSE are present alongside AM hyphae, see Hybrid

[X] No root /  
Background

Must exhibit the above features  
■ No cell walls are visible



|                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                       | <p><b>Percentage of the tile occupied:</b><br/>Less than 10% of tile includes cell walls</p> <p><b>Exemplary images:</b></p> <p><b>Other aspects to be aware of:</b><br/>May include bubbles, specks of dust, flecks of dirt, and extracellular hyphae</p>                                                                                                                                                                                                                                                                                                                                                                            |
| <p>[U] Unreadable</p> | <p>Designed for tiles that cannot be assessed by an expert.</p> <p><b>May exhibit following image features:</b><br/>Structures too blurry to assess, root too obscured by dirt to assess, stain too dark to see structures, too much extracellular hyphae to determine colonisation inside root, unconfirmed hyphal structures.</p> <p><b>Exemplary images:</b></p> <div data-bbox="632 1120 1158 1541">  </div> <div data-bbox="619 1632 1150 2056">  </div> |



**Other aspects to be aware of:**

If a tile is clear enough with context to be sure it falls into one of the other categories, some obscuring is acceptable. This is only for tiles that you are genuinely unsure of.

[H] Hybrid

Designed to accommodate cells with both AM colonisation and DSE. Must exhibit following image features:

- Those associated with AM+ **AND**
- Those associated with DSE

**Exemplary images:**





**Other aspects to be aware of:**

One may be dominant, but if both features are visible, it should still be included in this category.